ResearchGate

See discussions, stats, and author profiles for this publication at: https://www.researchgate.net/publication/10958047

Microstimulation of macaque V1 disrupts target selection: effects of stimulation polarity.

Article *in* Experimental Brain Research · February 2003

DOI: 10.1007/s00221-002-1312-5 · Source: PubMed

CITATIONS
25

READS
85

2 authors:

Edward J Tehovnik
Independent
78 PUBLICATIONS **2,863** CITATIONS

SEE PROFILE

Warren M Slocum
Massachusetts Institute of Technology
43 PUBLICATIONS **1,181** CITATIONS

SEE PROFILE

Some of the authors of this publication are also working on these related projects:

Project The Osh View project

Project Consciousness and Information Transfer View project

All content following this page was uploaded by Edward J Tehovnik on 17 May 2014.

The user has requested enhancement of the downloaded file.

Exp Brain Res (2003) 148:233–237
DOI 10.1007/s00221-002-1312-5

RESEARCH ARTICLE

Edward J. Tehovnik · Warren M. Slocum

Microstimulation of macaque V1 disrupts target selection: effects of stimulation polarity

Received: 4 July 2002 / Accepted: 4 October 2002 / Published online: 22 November 2002
© Springer-Verlag 2002

Abstract Microstimulation of the intermediate layers of V1 in rhesus monkeys disrupts target selection with saccadic eye movements. To study target selection, one visual target was presented in the receptive-field location of the stimulated neurons and a second target was presented outside this location. Microstimulation delivered with the appearance of the two targets decreased the probability that the monkey would select the target placed in the receptive-field location when intermediate layers of V1 were stimulated. This interference effect was more pronounced when anodal-first pulse pairs were used as compared to when cathodal-first pulse pairs were used. The superior effectiveness of anodal pulses suggests that the interference effect is due to activation of axonal terminals residing within intermediate V1.

Keywords Occipital cortex · Microstimulation · Cathodal pulses · Anodal pulses · Rhesus monkeys

Introduction

The striate cortex (area V1) mediates visual processes including target selection (Hubel and Wiesel 1968; Michael 1981; Ts'o and Gilbert 1988; Knierim and van Essen 1992; Kapadia et al. 1995; Rossi et al. 1996; Schiller and Tehovnik 2001, 2002; Tehovnik et al. 2002). The delivery of subthreshold microstimulation using cathodal-first pulse pairs to the intermediate layers of V1 in macaque monkeys interferes with the selection of visual targets located at the visual receptive field of the stimulated neurons (Schiller and Tehovnik 2001; Tehovnik et al. 2002). The cortical depth at which the interference effect is most pronounced about 0.8 mm below the cortical surface (Tehovnik et al. 2002). Anatomically, this depth is situated within layer IV of V1 (Peters and Sethares 1991), which receives visual inputs from the lateral geniculate nucleus (Hubel and Wiesel 1972). Given that V1 is a major conduit of visual information between the retina and higher cortical areas (Schiller et al. 1974), we have proposed that stimulation of intermediate V1 interrupts the flow of this information (Tehovnik et al. 2002). The purpose of the present study was to target the visual input fibres of V1 by using anodal-first pulse pairs. Anodal pulses are most effective at activating fibre terminals (Clendenin et al. 1974). We found that stimulation delivered to V1 disrupted target selection more when anode-first pulses were used than when cathode-first pulses were used.

E.J. Tehovnik (✉) · W.M. Slocum
Department of Brain and Cognitive Sciences,
Massachusetts Institute of Technology, E25–634, Cambridge,
MA 02139, USA
e-mail: tehovnik@mit.edu
Tel.: +1-617-2535792
Fax: +1-617-2538943

Materials and methods

One adult rhesus monkey (*Macaca mulatta*) was used. Throughout this study food was freely available. The monkey was deprived of water overnight before each day of experimental testing. After testing, he was allowed to drink until sated before being returned to the vivarium. An eye coil, head post, and recording well over V1 were implanted using standard methods (Tehovnik et al. 2002). Principles of animal care were followed according to NIH guidelines.

The monkey, with head fixed, faced a computer monitor positioned 57 cm away. All tasks started with the animal acquiring a fixation spot (6 min of arc in diameter). During this period, the animal was required to keep his eyes within a 0.5×0.5-degree window; otherwise the trial was aborted. Two types of tasks were performed. The first task was used to map the receptive-field location of V1 cells (Fig. 1A). Here the monkey was required to remain on the fixation spot as a high-contrast (>100% contrast) bar of light was swept across the visual field. After termination of the bar stimulus, the monkey generated a saccade to a target to obtain a reward of apple juice.

The second task, the paired-target task, required a monkey to remain on the fixation spot for 300 ms. One hundred milliseconds after termination of the fixation spot, two targets (each 12 min of arc in diameter at 100% contrast) were presented (Fig. 1B). One target was centred on the receptive-field location of the V1 cells under study and the remaining target occurred in the mirror position of the opposite hemifield. The targets were presented at various temporal offsets, with the receptive-field target occurring first on

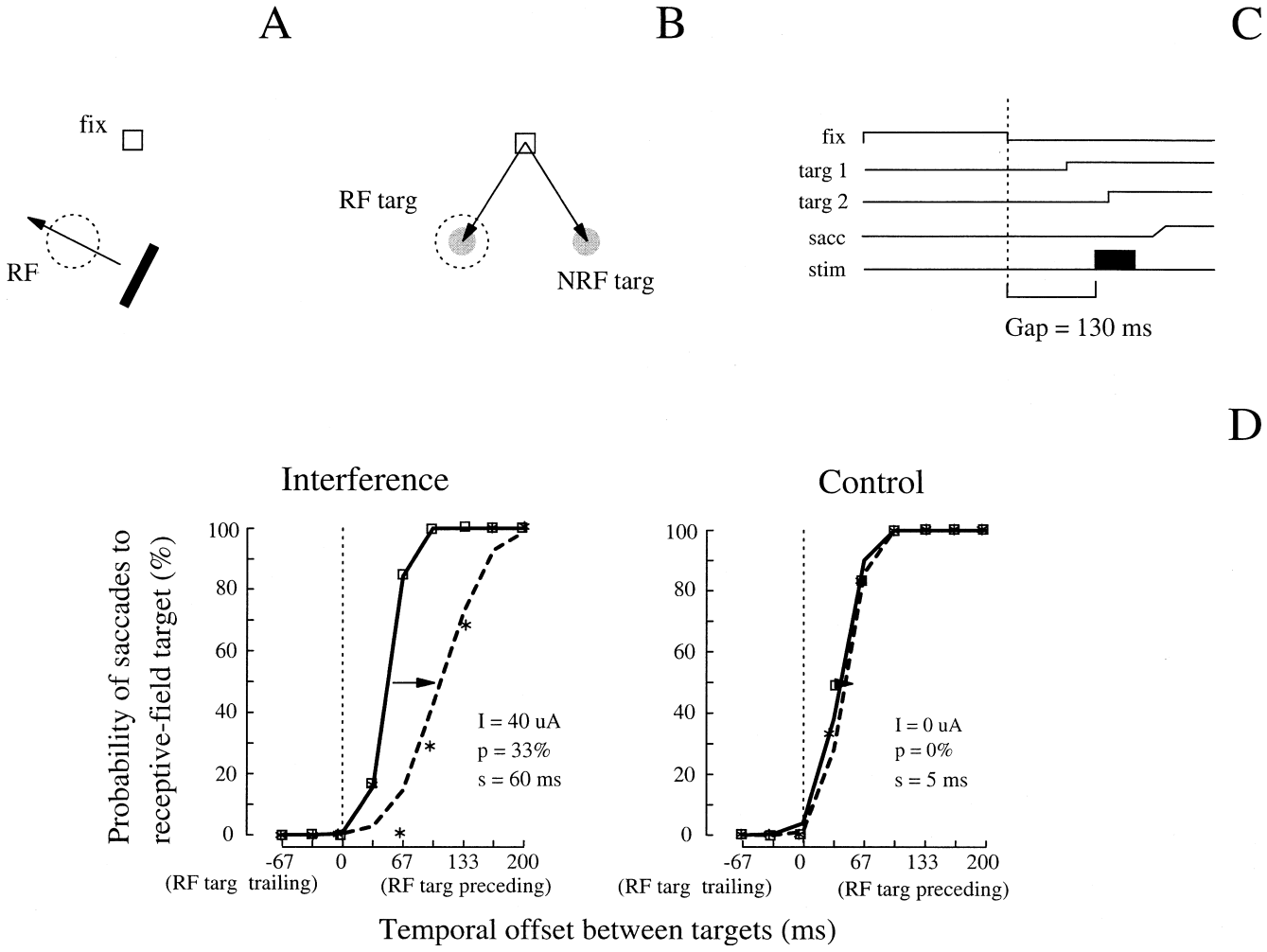

Fig. 1 Methods. **A** A monkey was required to fixate a spot (*fix*) as a bar stimulus was swept across the visual field to map the visual receptive-field (*RF*) of the units at the electrode tip. **B** Two targets spaced temporally were presented to the monkey. One target, the receptive-field target (*RF targ*), was positioned in the receptive-field location of the units at the electrode tip; the other target, the non-receptive-field target (*NRF targ*), was positioned outside of the receptive-field location in the mirror opposite hemifield from the receptive-field target. **C** A monkey was required to fixate a spot for 300 ms (*fix*). One hundred milliseconds after the termination of the fixation spot one of the paired targets was presented (*targ 1*). The second target of the pair (*targ 2*) was presented some time after the presentation of the first target or at the same time as the first target appeared. Electrical stimulation was delivered 30 ms after the presentation of the first target (*stim*). In order to obtain a juice reward, the monkey was required to generate a saccadic eye movement to one of the two targets within 300 ms following the onset of the initial target. **D** Shown is how the effect of electrical stimulation on the paired-target task was measured for a given stimulation site. The probability of saccades made to the target in the receptive field was plotted as a function of the temporal offset between targets for stimulation (*dashed curves*) versus non-stimulation (*solid curves*) trials. The data (depicted as *stars* for stimulation trials and *squares* for control trials) were fitted using Probit regression. Each data point is based on six trials. Negative values along the *x*-axis of each graph indicate that the receptive-field target was trailing the non-receptive field target (*RF targ trailing*); positive values along the *x*-axis indicate that the receptive-field target was preceding the non-receptive field target (*RF targ preceding*). When electrical stimulation interferes with target selection (*Interference*), the *dashed curve* is shifted rightward with respect to the *solid curve*. This indicates that the monkey selected the non-receptive-field target more often when the electrical stimulation was delivered as compared to when the stimulation was not delivered. The amount of curve shift was measured in milliseconds at the 50% saccade probability point to yield a displacement value (*s*) indicated by the *arrow*. Control experiments were conducted to determine the amount of curve shift in the absence of any electrical stimulation (*Control*). The current used (*I*) and the probability of evoking saccades on blank, non-target trials (*p*) are shown. Each probability value for saccades evocation is based on six trials

some trials, with the non-receptive-field target occurring first on other trials, and with both targets occurring simultaneously on remaining trials. The monkey was given a juice reward for generating a saccade to either target within 300 ms. On 50% of trials, electrical stimulation was delivered to V1 30 ms after the presentation of the first target of a pair (Fig. 1C). On some trials, no targets appeared after which a juice reward was delivered immediately. On 50% of such trials, electrical stimulation was delivered 130 ms after the termination of the fixation spot. These trials were used to determine the probability of evoking a saccade into the receptive-field location in the absence of visual targets.

A PDP 11/73 computer controlled the presentation of visual stimuli, the delivery of electrical stimulation, the display and

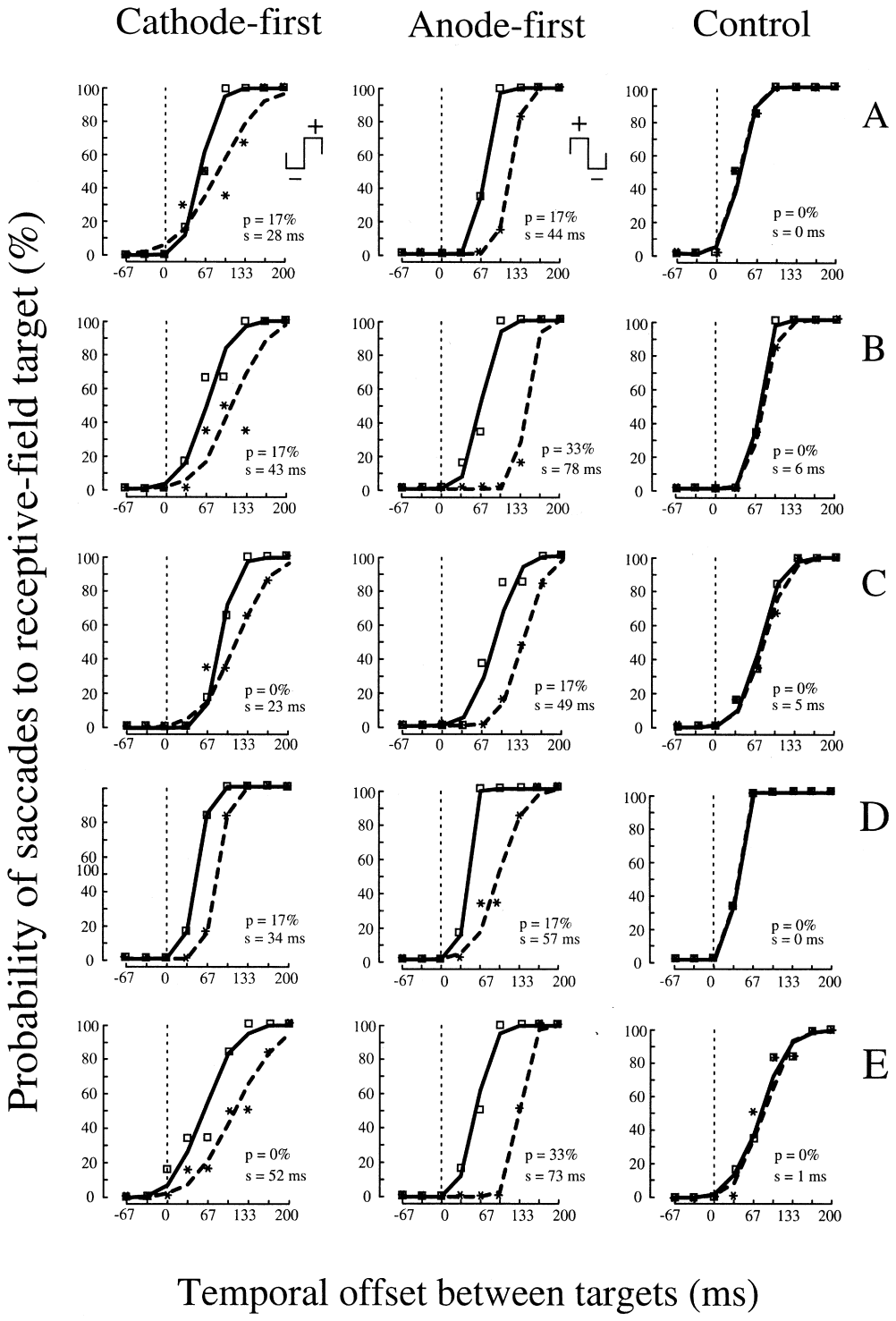

Fig. 2A–E Cathode-first versus anode-first induced interference. The probability of evoking saccades toward the receptive-field target is plotted as a function of the temporal offset between targets. Each row of plots, **A–E**, shows interference effects induced at a given site using cathode-first (*Cathode-first*) and anode-first (*Anode-first*) pulses followed by a non-stimulation control (*Control*). Icons representing the configuration of the stimulation pulses are shown in the *top two panels*. Within a panel, the *solid black curve* is the control and the *dashed curve* represents the effect of stimulation or the effect of dummy stimulation. For a given site **(A–E)**, the current level was fixed. All currents ranged from 40 to 50 μA. The probability of evoking saccades on blank, non-target trials (*p*), and the

collection of eye position (sampled at 200 Hz), and the delivery of juice.

A standard analysis was conducted on data obtained using the paired-target task. The probability of saccades made to the receptive-field target was plotted as a function of the temporal offset between targets (Fig. 1D). All data were fitted using Probit regression using SPSS. When the probability function for stimulation trials was shifted rightward with respect to the control function, this was designated as an interference shift (Fig. 1D, Interference). This indicated that the animal selected the target outside the receptive-field location of the stimulated neurons. Probability functions in which no stimulation was delivered on dummy trials were also determined (Fig. 1D, control).

Constant-current charge-balanced biphasic pulses were delivered to the brain via a monopolar electrode (glass-coated platinum-iridium) using a Grass S88 stimulator attached to a pair of constant-current stimulus isolation units (Grass PSIU6B). Biphasic pulses were used to reduce the chances of electrode polarization and tissue damage (Tehovnik 1996). Two pulse-pair configurations were used: (1) an anodal pulse followed by a cathodal pulse to produce

anode-first pulses and (2) a cathodal pulse followed by an anodal pulse to produce cathode-first pulses. Both pulses of a pair had the same amplitude and duration. Current was measured by the voltage drop across a 1,000-Ω resistor that was in series with the return lead of the stimulator. Pulses of 50 μA or less (at a 0.2-ms pulse duration) with pulse frequencies of 200 Hz and train durations of 100 ms were used. To increase the chances of evoking interference effects from V1, the currents were subthreshold, namely, those that evoked saccades on less than 50% of blank, non-target trials; and the electrode was positioned in the intermediate layers of V1 between 0 and 1.6 mm below the cortical surface (Tehovnik et al. 2002). Prior to stimulation, an electrode was introduced perpendicular to the cortical surface until action potentials were encountered. The action potentials were amplified (Bak A-1B) and filtered (Krohn-Hite 3750) and isolated with a window discriminator (Bak, DIS-1).

Results

All electrode penetrations made into V1 were made normal to the cortical surface. The penetrations contained cells with receptive-field centres varying between 2 and 3.6 degrees from the fovea and between 230 and 265 degrees of meridian. Whenever the stimulation evoked a saccadic eye movement, the saccade always terminated in the receptive-field location of the stimulated neurons (Schiller and Tehovnik 2001; Tehovnik et al. 2002).

The biphasic pulse-pair configuration, namely, whether the biphasic pulses started with a cathodal or anodal pulse, affected the magnitude of the interference effect observed during target selection. The probability of saccades toward the receptive-field target is plotted as a function of the temporal offset between targets for five stimulation sites in V1 located between 0.8 and 1.4 mm below the cortical surface (Fig. 2). The magnitude of the interference effect was greater when anode-first pulses were delivered (average curve shift = 60 ms) than when cathode-first pulses were delivered (average curve shift = 36 ms; $t_{(4)}$=7.7, P<0.01). The probability of evoking saccades on blank, non-target trials increased somewhat for the anode-first condition (i.e., cathodal: 10%; anodal: 23%), but this difference was not significant (P>0.05). Finally, the magnitude of the interference effect was greater when cathode-first pulses were delivered (average curve shift = 36 ms) than when no pulses were delivered (average curve shift = 2 ms; $t_{(4)}$=6.2, P<0.01).

Discussion

We observed that anodal-first stimulation was more effective at inducing interference than was cathodal-first stimulation. Clendenin et al. (1974) found that the current threshold for activating cerebellar mossy fibre axon terminals was on average 36% lower when anodal pulses were delivered as compared to when cathodal pulses were delivered.[1] Why are anodal pulses more effective at activating axon terminals? According to Ranck (1975), effective stimulation of neural tissue induces an outward current at the initial segment or node of Ranvier, thereby triggering an action potential. When cathodal current is delivered adjacent to a neural element an outward current is induced causing the membrane to be depolarized, whereas when an anodal current is delivered an inward current is induced causing the membrane to be hyperpolarized. It is for this reason that cathodal pulses are more effective than anodal pulses at activating axons (Porter 1963; Stoney et al. 1968; Armstrong et al. 1973) and it is because of this that cathodal pulses are routinely used for brain stimulation (Tehovnik 1996). When an anodal current is delivered to an axon terminal an inward current is produced at the terminal while an outward current occurs upstream at the first node of Ranvier. This outward current activates the neuron. When cathodal current is delivered to an axon terminal, an outward current is produced at the terminals and an inward current is produced at the first node of Ranvier. The inward current at the first node of Ranvier hyperpolarizes the membrane, thereby making the neuron less excitable.

What the foregoing implies is that anodal pulses are more effective at activating terminals than they are at activating axons. At the input layers of V1 there are at least two terminal systems: a terminal system originating from the lateral geniculate nucleus (Hubel and Wiesel 1972) and a second system intrinsic to V1 comprising GABAergic interneurons (Shaw and Cynader 1986; Fitzpatrick et al. 1987). At this point, it is uncertain whether the stimulation is disrupting target selection by stimulating fibres from the lateral geniculate nucleus or by activating the GABAergic fibres or by stimulating both systems. It is noteworthy that GABAergic agonists and antagonists injected into V1 are effective at disrupting target selection (Schiller and Tehovnik 2002).

In a previous report (Tehovnik et al. 2002), we postulated that the delivery of electrical stimulation to the intermediate layers of V1 disrupts the flow of visual information between the retina and higher cortical areas. Several observations support this notion. First, stimulation near or at the visual input layers interferes with the selection of visual targets situated at the receptive-field location of the activated neurons (current study; Schiller and Tehovnik 2001; Tehovnik et al. 2002). Second, anodal-first pulse pairs induce interference more readily than do cathodal-first pulse pairs, suggesting that the interference effect is due to stimulation of fibre terminals located within the visual input layers of V1 (current study). Third, as the electrode is situated more distant from V1, the chances of observing interference diminishes. Interference is most notable for stimulations of V1 and V2, somewhat less notable for stimulation of the lateral intraparietal area, and not at all apparent for stimulations of the frontal and medial eye fields (Schiller and Tehovnik 2001).

Perceptual interference induced by presenting two visual stimuli in rapid succession to the same parts of the visual field is a form of visual masking (Breitmeyer 1984). It is conceivable that the introduction of electrical

[1] Monophasic pulses were used.

stimulation to neurons in V1 30 ms after the onset of a visual target that falls in the receptive field of those neurons produces such a masking effect. Schiller (1969) has reported that unit activity in V1 is affected by visual masking. We have yet to vary the time between target onset and stimulation delivery using a range of currents and train durations to ascertain whether our interference effect exhibits properties similar to that of visual masking.

Acknowledgements This work was supported by NIH EY08502 to P.H. Schiller. We would like to thank Ms. Christina E. Carvey for helping in the preparation of the manuscript.

References

Armstrong DM, Harvey RJ, Schild RF (1973) The spatial organization of climbing fibres branching in the cat cerebellum. Exp Brain Res 18:40–58

Breitmeyer BG (1984) Visual masking: an integrative approach. Oxford University Press, New York

Clendenin M, Ekerot C-F, Oscarsson O, Rosen I (1974) The lateral reticular nucleus in the cat. I. Mossy fibre distribution in cerebellar cortex. Exp Brain Res 21:473–486

Fitzpatrick D, Lund JS, Schmechel DE, Towles AC (1987) Distribution of GABAergic neurons and axon terminals in the macaque striate cortex. J Comp Neurol 264:73–91

Hubel DH, Wiesel TN (1968) Receptive fields and functional architecture of monkey striate cortex. J Physiol (Lond) 195:215–243

Hubel DH, Wiesel TN (1972) Laminar and columnar distribution of geniculocortical fibres in the macaque monkey. J Comp Neurol 146:421–450

Kapadia MK, Ito M, Gilbert DC, Westheimer G (1995) Improvement in visual sensitivity by changes in local context: parallel studies in human observers and in V1 of alert monkey. Neuron 15:843–856

Knierim JJ, van Essen DC (1992) Neuronal responses to static texture patterns in area V1 of the alert macaque monkey. J Neurophysiol 67:961–980

Michael RC (1981) Columnar organization of color cells in monkey's striate cortex. J Neurophysiol 46:587–604

Peters A, Sethares C (1991) Organization of pyramidal neurons in area 17 of monkey visual cortex. J Comp Neurol 306:1–23

Porter R (1963) Focal stimulation of hypoglossal neurones in the cat. J Physiol (Lond) 169:630–640

Ranck JB (1975) Which elements are excited in electrical stimulation of mammalian central nervous system: a review. Brain Res 98:417–440

Rossi AF, Rittenhouse CD, Paradiso MA (1996) The representation of brightness in the primary visual cortex. Science 260:995–997

Schiller PH (1969) Behavioral and electrophysiological studies of visual masking. In: Leibovic KN (ed) Information processing in the nervous system. Springer, New York, pp 141–165

Schiller PH, Tehovnik EJ (2001) Look and see: how the brain moves the eyes. Prog Brain Res 134:127–142

Schiller PH, Tehovnik EJ (2002) The role of cortical inhibitory circuits in target selection with saccadic eye movements. Vis Sci Soc Abstr 2:200

Schiller PH, Stryker M, Cyander M, Berman N (1974) The response characteristics of single cells in the monkey superior colliculus following ablation or cooling of visual cortex. J Neurophysiol 37:181–194

Shaw C, Cynader M (1986) Laminar distribution of receptors in monkey (*Macaca fascicularis*) geniculostriate system. J Comp Neurol 248:301–312

Stoney SD, Thompson WD, Asanuma H (1968) Excitation of pyramidal tract cells by intracortical stimulation: effective extent of stimulating current. J Neurophysiol 31:659–669

Tehovnik EJ (1996) Electrical stimulation of neural tissue to evoke behavioural responses. J Neurosci Methods 65:1–17

Tehovnik EJ, Slocum WM, Schiller PH (2002) Differential effects of laminar stimulation of V1 on target selection by macaque monkeys. Eur J Neurosci 16:751–760

Ts'o DY, Gilbert CD (1988) The organization of chromatic and spatial interactions in the primate striate cortex. J Neurosci 8:1712–1727

View publication stats